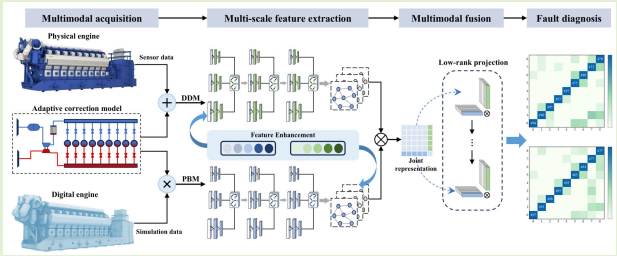
An Enhanced Digital Twin-Driven Fault Diagnosis Approach Based on Deep Multimodal Information Fusion for Marine Machinery

Jiawen Sun, Hongxiang Ren, Yaxin Dong, Boxiang Zhang, Shaoyang Qiu, and Yi Zhou

***Abstract*—Condition monitoring and fault diagnosis of marine machinery are essential for ensuring the safety and stability of offshore operations. Traditional approaches, which typically rely solely on either physical mechanisms or large volumes of data, often struggle to achieve both high accuracy and adaptability to component degradation. To address this issue, this article proposes a novel digital twin framework based on deep multimodal information fusion (DMIF-DT) that integrates the complementary strengths of a physics-based model (PBM) and a data-driven model (DDM). An attention-enhanced multiscale convolutional network is developed to extract features from sensor measurements and calibrated simulation data. A low-rank fusion strategy is then employed to map the two modalities into a unified high-dimensional space for effective integration between the PBM and the DDM. Furthermore, a performance-degradation adaptive correction module, based on sensor measurements and probabilistic distribution features, is used to map health factors and feed them back to the PBM for component-level correction, thereby supporting periodic model updates. Experimental results show that the proposed DMIF-DT framework significantly reduces parameter prediction errors and achieves an average fault diagnosis accuracy of 98.02% under diverse operating conditions. By uniquely integrating a performance-degradation adaptive correction module with attention-enhanced multimodal fusion, the framework bridges physics-based and data-driven insights, providing a new approach for the intelligent health management of marine machinery.**



***Index Terms*— Adaptive correction, digital twin (DT), fault diagnosis, marine machinery, multimodal fusion.**

I. Introduction

THE intelligent management of marine machinery systems is crucial for ensuring the safety and stability of maritime operations [1]. As operational time under harsh conditions increases, key components of marine machinery undergo progressive performance degradation. This degradation can lead to reduced component functionality, fault occurrences, or even system failures. Therefore, it is of great significance to develop an accurate, reliable, and efficient fault detection and diagnosis method, which helps evaluate the health of machinery systems during voyages and supports effective maintenance decisions when docked.

Fault detection and diagnosis technology for marine machinery refers to the capability to identify and localize faults down to replaceable components or specific subsystems [2]. Timely and reliable diagnosis provides strong support for optimizing maintenance decisions and extending the service life of machinery, thereby reducing maintenance costs and improving system operational efficiency [3]. The two main techniques for marine machinery fault diagnosis are model-based methods and data-driven methods [4]. Model-based methods rely on prior knowledge of the structure or internal mechanisms of the target system to enable the representation and analysis of its operating states [5]. Milioulis et al. [6] developed a functional model of a marine liquefied natural gas (LNG) fuel supply system and identified critical faults by integrating failure mode, effects, and criticality analysis (FMECA) with fault tree analysis (FTA). In general, function-based modeling methods emphasize fault path reasoning and are well suited for system-level failure analysis. However, they depend on a predefined fault mode library and exhibit limitations in identifying performance degradation and

Received 26 November 2025; accepted 12 December 2025. Date of publication 22 December 2025; date of current version 2 February 2026. This work was supported in part by Guangxi Science and Technology Major Program under Grant GUIKE AA23062053, in part by Guangxi Key Research and Development Plan under Grant GUIKE AB22080106, in part by Dalian Science and Technology Innovation Fund Project under Grant 2022JJ12GX035, and in part by the Fundamental Research Funds for the Central Universities under Grant 3132025231. The associate editor coordinating the review of this article and approving it for publication was Dr. Xu Cheng. *(Corresponding author: Hongxiang Ren.)*

The authors are with the College of Navigation and the College of Artificial Intelligence, Dalian Maritime University, Dalian 116026, China, and also with CNOOC Energy Technology and Services-Oil Production Services Company, Tianjin 300452, China (e-mail: dmu_rhx@163.com).

Digital Object Identifier 10.1109/JSEN.2025.3644934

1558-1748 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

coupled faults. Physics-based models (PBMs) constructed from first principles are typically more accurate and capable of effectively capturing the nonlinear operational behavior of machinery [7]. For instance, Wang et al. [8] developed a PBM of a two-stroke low-speed marine diesel engine using MATLAB/Simulink to simulate typical fault scenarios, thereby supporting fault identification and health assessment. Benefiting from modular simulation strategies, PBMs offer strong interpretability. However, one key prerequisite for implementing PBMs is access to precise component characteristics, which are often derived under ideal design conditions. When the actual system deviates from the design point, PBM reliability and accuracy may decline. Moreover, as the operational life of ship systems increases, component performance inevitably degrades, making it difficult for PBMs to maintain high-fidelity simulation of real-world behavior. Therefore, the dynamic mismatch between PBMs and the physical system remains a significant challenge for current model-based approaches [9].

On the other hand, data-driven models (DDMs) use collected data and prior experience to extract high-dimensional features representing machinery health and construct fault identification models [10]. Numerous studies on DDMs have focused on modeling marine systems using machine learning techniques, including ensemble learning [11], support vector machines [12], extreme learning machines [13], and various neural network architectures featuring efficient nonlinear multilayer modeling and adaptability [14], [15]. Although these methods do not require an in-depth understanding of the underlying physical mechanisms, they also present certain limitations. In practice, the imbalance and incompleteness of measurement data, together with the scarcity of labeled samples, make it challenging to obtain high-quality training datasets [16]. Moreover, inevitable noise and redundancy in the acquired data may distort feature representations and thus degrade the diagnostic performance of DDMs. To address these limitations, recent studies have proposed a variety of more advanced DDMs. For example, Huang et al. [17] developed a deep adversarial capsule network (DACN) that combines capsule networks with adversarial domain generalization for compound fault diagnosis under unseen conditions. Zhang et al. [18] proposed a self-supervised graph feature enhancement and scale attention-driven framework (EG-SAGCN) to improve diagnosis with limited labeled data. In addition, to address the challenges of nonstationary and imbalanced conditions in machinery fault diagnosis, Xu et al. [19] designed a domain-constrained cascadic multireceptive learning network (DC-CMLN) integrating multiscale feature extraction and domain-constrained label adjustment. Li et al. [20] presented a sensor-aware capsule neural network that integrates interpretable multisensory fusion with uncertainty-aware estimation for reliable remaining useful life prediction. The above advanced DDMs have demonstrated promising results in industrial prognostics and health management (PHM) tasks, providing valuable insights for developing more robust and generalizable diagnostic frameworks. However, their application in marine machinery fault diagnosis still faces challenges, including the limited exploitation of physical knowledge and inadequate adaptability to performance degradation.

Since the concept of the digital twin (DT) was proposed, it has attracted widespread interest [21]. As an innovative approach, the DT enables high-fidelity system modeling and supports condition awareness and management throughout the entire equipment lifecycle, which provides a more effective solution for fault prediction and health management of marine machinery. The core focus of the DT lies in the interaction between the physical and virtual spaces [22]. Multisource information from complex systems—such as sensor data and simulation model outputs—is integrated and transmitted to the virtual space to construct the DT model. Subsequently, the DT model enables functions such as condition monitoring and fault diagnosis. Finally, feedback information is sent back to the physical space to support optimization and decision-making.

As mentioned above, PBMs and DDMs represent two distinct paradigms in marine machinery modeling—mechanism-based and data-driven models, respectively. One of the key challenges in constructing DT systems is the effective integration of these two approaches [23]. Currently, DT applications in marine machinery remain in an early stage of development, and research on model construction is still relatively limited. Hautala et al. [24] developed a DT system for marine engines based on a 1-D thermodynamic model, enhancing real-time simulation capability through optimization and simplification strategies. However, this study did not consider degradation adaptive modeling, suggesting that the DT model may have difficulty adapting when component performance declines due to wear, corrosion, or other aging effects during vessel operation. To address this issue, Jeon and Theotokatos [9] proposed a trustworthiness assurance framework for marine engine DTs that incorporates residual analysis and credibility evaluation to mitigate model mismatches arising from degradation, thereby improving the engineering adaptability of the DT system. Li and Brennan [25] proposed a DT framework for virtual maintenance of complex marine equipment, integrating structural modeling with operational data monitoring for predictive maintenance and fault warning. Similarly, Mousavi et al. [26] presented a DT framework combining finite element-based physical modeling with Bayesian uncertainty reasoning to detect damage and update system states under uncertain loading conditions. These methods further underscore the important role of DDMs in DT model development. Notably, DDMs are proficient in learning patterns and trends from data, making them particularly advantageous for modeling component degradation [27]. However, for complex marine machinery such as ship engines and propeller propulsion systems, DT models that effectively integrate PBMs and DDMs remain relatively underexplored in fault diagnosis and health management applications, especially when performance degradation is considered.

Similar to how humans perceive the external environment through multiple sensory channels, such as vision and hearing, multimodal learning can integrate multisource heterogeneous information from complex systems by constructing joint representations [28]. In DT systems, the PBM typically offers high

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

modeling accuracy and interpretability but is not well suited for scenarios involving significant performance degradation [29], whereas the DDM excels at capturing data patterns and trends. Although recent hybrid DT and physics-informed learning approaches have combined these two paradigms to enhance diagnostic robustness, their information interaction and fusion remain insufficient. For instance, most existing frameworks adopt a unidirectional mapping strategy that either uses measurement data to correct PBM simulation outputs or employs PBM-generated data to augment DDM training. Therefore, it remains necessary to explore new strategies for the deep integration of PBM and DDM models, enabling feedback-based interaction and adaptive model updating within the DT framework.

Given the above, a novel DT approach based on deep multimodal information fusion (DMIF-DT) is proposed for fault diagnosis of marine machinery. Compared with traditional hybrid DT methods that typically adopt a one-way mapping between physical and data models, the proposed framework establishes a feedback-enabled two-way interaction between the PBM and DDM, forming a closed-loop information exchange that enhances model consistency and integration effectiveness. Specifically, the PBM is constructed using a 1-D high-fidelity thermodynamic model, whereas the DDM utilizes an attention-enhanced multiscale feature fusion (AMSF) method to extract relevant data features. The two modalities are then integrated using a low-rank fusion strategy to form a joint representation. Furthermore, to allow the DT model to adapt to component performance degradation, a performance-degradation correction (PDC) module is introduced. This module adjusts the model using adaptive factors computed from sensor data and probability distribution outcomes, enabling better alignment with real-world measurements. Overall, the main contributions of this article are as follows.

1) A deep multimodal information fusion DT framework is developed, featuring a feedback-enabled two-way interaction between the PBM and DDM to achieve more effective integration.
2) A PDC mechanism, driven by Kullback–Leibler (KL)-divergence-based health indicators (HIs), is proposed to adaptively refine PBM parameters and maintain long-term consistency with real operating conditions.
3) An attention-enhanced multiscale feature (MSF) extraction and low-rank fusion network is adopted as an effective integration scheme to improve feature discriminability and capture high-order correlations across modalities.
4) Compared with conventional model-based and data-driven methods, the proposed DMIF-DT framework demonstrates higher diagnostic accuracy and robustness across diverse operating conditions.

The rest of this article is organized as follows. Section II presents the basic theories used in constructing the PBM and DDM. Section III describes the architecture and implementation of the proposed DT framework in detail. Section IV systematically validates the proposed approach through a series of comparative experiments. Finally, Section V draws conclusions and outlines future research directions.

TABLE I
TECHNICAL SPECIFICATIONS OF WARTSILA 9L34DF DUAL-FUEL ENGINE

Technical specifications	Details
Number of cylinders	9
Stroke	4
Power	4050 kw
Engine speed	750 r/min
Bore	340 mm
Piston stroke	400 mm
Firing order	1-7-4-2-8-6-3-9-5

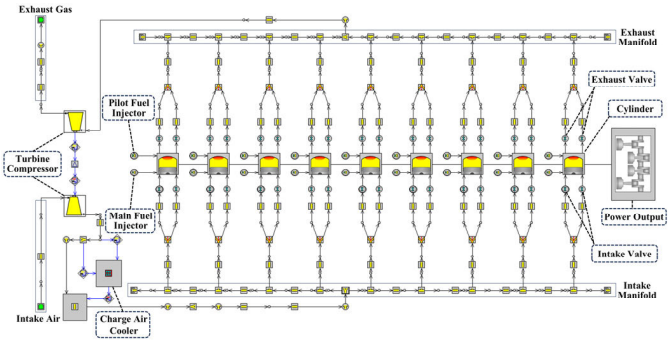


Fig. 1. Simulation model of the marine engine built using GT-SUITE.

II. THEORETICAL BACKGROUND AND MODELING

A. Establishment of the Marine Engine Simulation Model

As a core system of marine machinery, the operational status of the marine engine directly affects the vessel's propulsion performance and overall efficiency. Therefore, a simulation model of a marine engine is developed in this study to serve as the original PBM within the DT framework. The model is based on a Wartsila 9L34DF dual-fuel, medium-speed, turbocharged diesel engine, which functions as the generator prime mover in an electrically propelled ship. The detailed specifications of the engine are listed in Table I. Based on these technical specifications, the simulation model is constructed using GT-SUITE, as shown in Fig. 1. For a detailed description of the modeling process using GT-SUITE for marine engines, refer to [30].

As shown in Fig. 1, the original PBM is developed using the standard component library available in GT-SUITE. Specifically, the heat transfer and combustion processes in the cylinder (CYL) are modeled using the Woschni equation and the Wiebe function, respectively. The intake and exhaust piping systems are simulated using 1-D methods to capture gas flow characteristics. In addition, the turbocharger (TC) unit consists of a compressor and a turbine. The compressor is positioned between the ambient environment and the charge air cooler (CAC), while the turbine is located between the exhaust manifold and the exhaust outlet. The input and output ports of each component are connected according to mass flow balance principles and the ideal gas law, allowing serial coupling

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

among components and forming a complete physics-based simulation model.

B. Convolutional Neural Network

The convolutional neural networks (CNNs), widely used in computer vision, have also shown strong potential for processing time-series data. Owing to its sparse connectivity and weight sharing, CNN can efficiently extract deep features from high-dimensional data [31]. In this study, the collected marine system parameters are 1-D time-series data; hence, feature extraction is performed using 1-D convolution. In the convolution process, the kernel functions as a filter that extracts features. The key formula associated with this process is as follows:

$$X_j^{(l+1)}(\tau) = \delta\left(b_j^l + \sum_{f=1}^{F^l} K_{jf}^l(\tau) \otimes X_f^l(\tau)\right) \tag{1}$$

where $X_j^{(l+1)}(\tau)$ denotes the activation of the jth feature map at time step τ in the $(l+1)$th layer, representing the output after the convolution operation. $\delta(\cdot)$ denotes the activation function, for which the ReLU function is adopted in this study. b_j^l is the bias term, which adjusts the output of the convolution calculation. F^l denotes the number of input feature maps in the lth layer. K_{jf}^l denotes the convolution kernel that connects the fth input feature map to the jth output feature map in layer l. The operator $\otimes$ represents the 1-D convolution. $X_f^l(\tau)$ denotes the value of the fth input feature map at time step τ in layer l, i.e., the input to the current layer.

The pooling layer processes the feature maps produced by the convolutional layer, performing downsampling to reduce computational complexity while preserving local structural information. The average pooling operation, for example, is computed as follows:

$$X_{j+1,\text{pool}}^l = \frac{1}{|\Omega_t|} \sum_{s \in \Omega_t} X_j^l(s) \tag{2}$$

where $X_{j+1,\text{pool}}^l$ is the output of the pooling operation at layer l. Ω_t is the index set of the tth pooling window, defined by $\Omega_t = \{s|(t-1)r \le s < (t-1)r + r\}$, with r denoting the kernel size. While convolutional layers focus on extracting deep hierarchical features, pooling layers reduce feature dimensionality to facilitate efficient and robust sequence modeling.

Convolution and pooling layers are alternately applied to extract hierarchical features, which are then passed to the fully connected (FC) layer for further processing and output generation. In addition, to accelerate computation and reduce dependence on network initialization, a batch normalization layer is inserted between the convolutional layer and the ReLU activation function. The computation of the batch normalization layer is defined in the following equation:

$$\bar{x}_j^{(k)} = \frac{x_j^{(k)} - E\left[x^{(k)}\right]}{\sqrt{\text{Var}\left[x^{(k)}\right]}} \tag{3}$$

where $x_j^{(k)}$ denotes the jth activation value in the kth batch; $E[x^{(k)}]$ and $\text{Var}[x^{(k)}]$ represent the mean and variance of the current batch, respectively; and $\bar{x}_j^{(k)}$ is the normalized activation value.

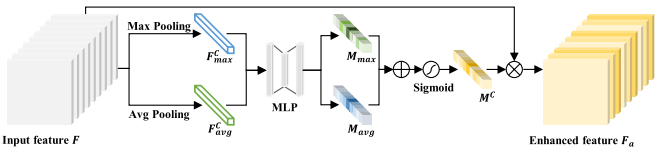


Fig. 2. Architecture of the CAM.

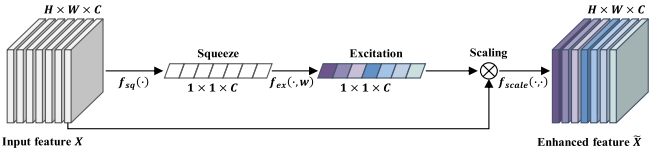


Fig. 3. Architecture of the SE module.

C. Attention Mechanism

The attention mechanism in deep learning is inspired by the ability of the human visual system to selectively focus on key targets or local regions during information processing [32]. In sequence modeling tasks, attention enables the model to capture the sensitivity of features across multiple dimensions. Its core idea is to assign importance weights to input elements so that the model focuses on the most relevant information for the task. This mechanism not only improves the model's ability to process complex data but also enhances its interpretability and computational efficiency [33]. In this study, the channel-attention module (CAM) and the squeeze-and-excitation (SE) module are introduced as fundamental building blocks.

The architecture of the CAM is shown in Fig. 2. In this module, the input feature map F is first compressed along the spatial dimensions using both max pooling and average pooling operations to facilitate the learning of channel-specific features [34]. The resulting features are subsequently passed through a shared multilayer perceptron (MLP) to generate corresponding weight vectors. Finally, a sigmoid activation function is applied to normalize the weights, producing the channel-attention map M^C. This process can be expressed as

$$\begin{aligned} M^C &= \sigma\left\{\text{MLP}\left[\text{Maxpool}(F)\right] + \text{MLP}\left[\text{Avgpool}(F)\right]\right\} \\ &= \sigma\left\{W_1\left[W_0\left(F_{\max}^C\right)\right] + W_1\left[W_0\left(F_{\text{avg}}^C\right)\right]\right\} \end{aligned} \tag{4}$$

where $M^C \in \mathbb{R}^{C\times1\times1}$ represents the channel-attention map. $F_{\max}^C \in \mathbb{R}^{1\times1\times C}$ and $F_{\text{avg}}^C \in \mathbb{R}^{1\times1\times C}$ are the max- and average-pooled features, respectively. σ denotes the sigmoid activation function. $W_0 \in \mathbb{R}^{C/r\times C}$ and $W_1 \in \mathbb{R}^{C\times C/r}$ are the weights of the MLP.

The SE module aims to enhance the representational quality of convolutional features by explicitly modeling interchannel dependencies, as illustrated in Fig. 3. It operates through two key stages: squeeze and excitation [35]. First, the input feature $X = [x_1, x_2, \ldots, x_c]$ undergoes a squeeze operation f_{sq}, which performs global information compression over the spatial dimensions to generate a compact descriptor for

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

each channel. Then, the excitation operation f_{ex}, implemented using two FC layers, models channel-wise dependencies and generates attention weights via a sigmoid activation function. Finally, a scaling operation f_{scale} is applied to rescale the original features according to the computed attention, producing attention-enhanced output features. This process is formulated as

$$z_c = f_{\text{sq}}(x_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j) \tag{5}$$

$$s_c = f_{\text{ex}}(z_c, W) = \sigma[W_2 \delta(W_1 z_c)] \tag{6}$$

$$\widetilde{x_c} = f_{\text{scale}}(x_c, s_c) = s_c \cdot x_c \tag{7}$$

where x_c denotes the feature value at location (i, j) in channel c. z_c is the global descriptor of channel c. s_c represents the attention weight generated by the excitation module. W_1 and W_2 are the weight matrices of the two linear layers. σ and δ denote the ReLU and sigmoid activation functions, respectively. $\widetilde{x_c}$ is the final attention-weighted output feature.

III. Methodology

A. Overall Framework of the Proposed DT

The proposed DT framework for marine machinery comprises the PBM, DDM, multimodal fusion, and functional modules, as illustrated in Fig. 4. This framework aims to construct a virtual DT model that closely reflects the actual operating behavior of a marine engine by integrating mechanism-based modeling with data-driven approaches, thereby supporting accurate fault diagnosis and degradation prediction.

The PBM performs simulation modeling by describing the physical and thermodynamic relationships among internal engine components, with its output serving as the mechanism-based modality. To address the inevitable performance degradation of engine components, a PDC module is employed to construct HIs that characterize degradation trends. These indicators are used to periodically refine key parameters of the PBM, maintaining the long-term accuracy and reliability of the simulation model.

The DDM is constructed using sensor data collected during actual engine operation to uncover latent patterns in the engine state data. Specifically, an AMSF method is designed to facilitate deep feature extraction and interaction. By enhancing the representation of degradation-related features under varying operating conditions, the AMSF improves the adaptability and robustness of the DT framework. Details of the AMSF method are provided in Section III-C.

A low-rank multimodal fusion (LMF) model is adopted to integrate PBM and DDM modalities, generating a joint representation in a high-dimensional feature space. Compared with single-modality information, the fused representation provides greater expressive capacity and improved diagnostic sensitivity. Based on this integrated representation, the DT performs condition monitoring and fault diagnosis for the marine engine. Diagnostic results and degradation information are periodically fed back through the PDC mechanism to refine PBM parameters.

B. Adaptive PDC for PBM

Performance degradation of marine machinery components overextended service periods is inevitable. This degradation causes key parameters to drift from their initial baselines and reduces PBM accuracy within the DT system. To address this issue, a PDC module is developed by integrating historical monitoring data and model knowledge. The module employs KL divergence to capture distributional differences in representative operating conditions across historical voyages. Using these differences, HIs are constructed to track degradation trends and enable periodic correction of key PBM components, thereby maintaining long-term model accuracy.

The overall structure of the PDC module is shown in Fig. 5, which consists of three blocks: KL divergence calculation, HI construction, and PBM update. First, the KL divergence calculation block estimates probability distributions of component performance parameters under consistent operating conditions using data from multiple voyages. For a representative condition c, performance feature distributions from two voyages are denoted as $P_c^{(i)}$ and $P_c^{(j)}$. To obtain these distributions, kernel density estimation (KDE) with a Gaussian kernel is applied to measurement samples under the healthy baseline and the current operating state. The bandwidth h follows Silverman's rule of thumb:

$$h = 1.06 \sigma n^{-1/5} \tag{8}$$

where σ and n denote the sample standard deviation and the sample size, respectively. The KL divergence between the two distributions is defined as

$$D_{\text{KL}}\left(P_c^{(i)} \parallel P_c^{(j)}\right) = \sum_{x \in X} P_c^{(i)}(x) \log \frac{P_c^{(i)}(x)}{P_c^{(j)}(x)}. \tag{9}$$

This process is repeated across multiple voyages to construct KL divergence vectors that capture changes in component conditions. Next, the HI construction block takes both the KL divergence vector and the current state signal S_{real} of the engine as input, and uses an MLP to predict a set of HIs

$$\text{HIs} = \mathcal{F}_\theta(S_{\text{real}}, D_{\text{KL}}) \tag{10}$$

where HIs $\in \mathbb{R}^n$ denotes the predicted values of n HIs, each corresponding to a key PBM component. $\mathcal{F}_\theta$ is the MLP model parameterized by θ. These indicators are not only highly sensitive to degradation trends but also physically interpretable.

Finally, the PBM update block performs adaptive correction of key model parameters based on the predicted HIs. Each degraded component is assigned a correction coefficient derived from its HI, which adjusts the associated PBM parameters through a bounded updating rule. Specifically, parameter variations are scaled by a predefined adaptation rate to control the correction amplitude and clipped within physically reasonable limits to prevent abrupt or nonphysical changes. Through this process, the PBM continuously tracks the evolving performance of the engine and remains consistent with actual engine conditions.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

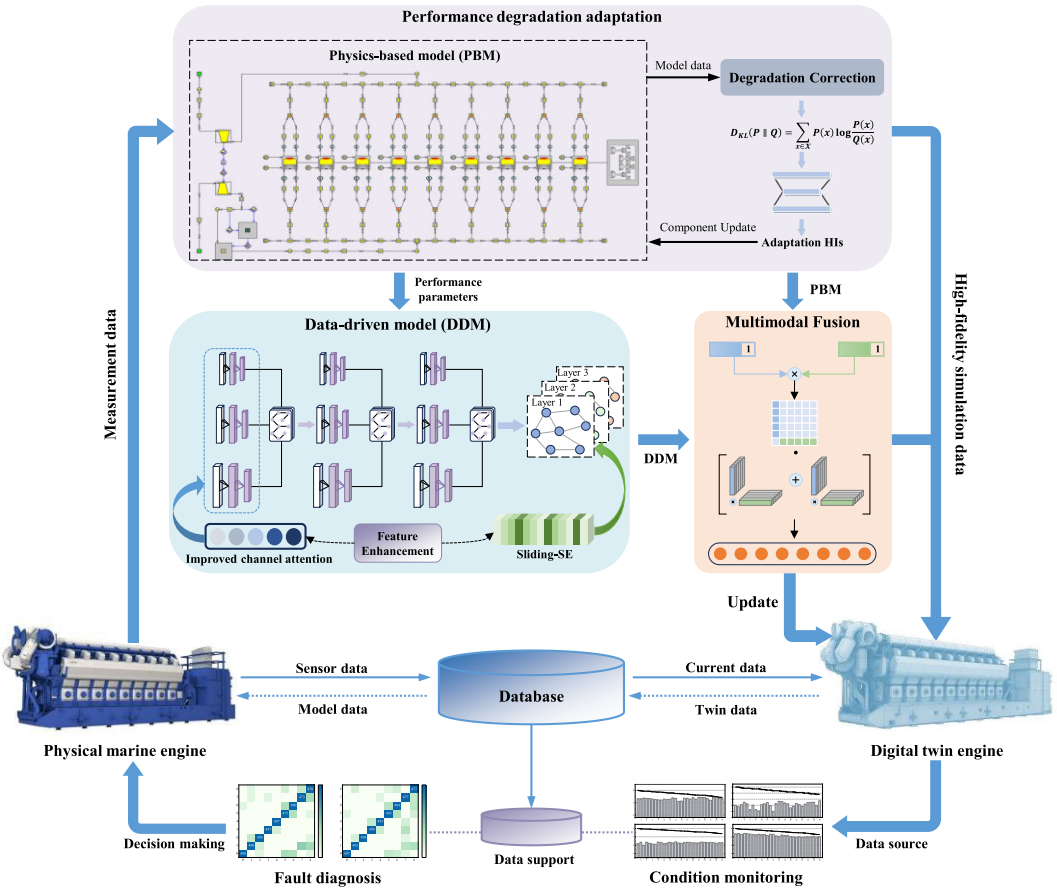


Fig. 4. Overall framework of the proposed DT for marine machinery.

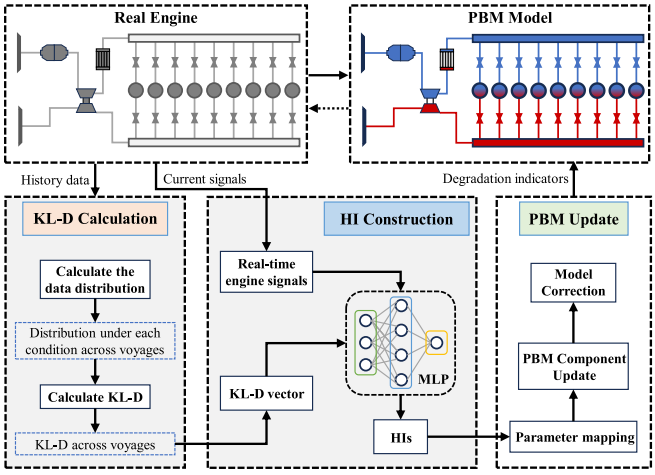


Fig. 5. PDC module for PBM updating under performance degradation.

C. Attention-Enhanced MSF Fusion

In this section, AMSF is developed to enhance the capability of the DDM in extracting critical fault features. It comprises an MSF extraction module, an improved CAM (ICAM), and an SE module based on sliding-window pooling (Sliding-SE).

1) MSF Extraction and Fusion: Effective feature extraction from diverse fault signals is essential for accurate diagnosis [36]. To overcome the limitations of single-scale CNNs, this study designs an MSF extraction and fusion module. Specifically, two improvements are introduced to the 1-D CNN. First, multiple convolutional kernels with different scales are applied in parallel to extract features from the input data, providing stronger generalization capabilities compared to single-scale convolution. Second, a cross-level feature fusion mechanism is incorporated to reduce potential information loss in deep networks. The fused features retain more detailed degradation information of mechanical components and offer a more comprehensive representation of system condition variations.

The architecture of MSF is shown in Fig. 6. After the initial convolutional, three identically structured MSF extraction blocks are stacked to iteratively extract and reconstruct meaningful intrinsic features from the input samples. In each block, kernels of different sizes ($w_1 \neq w_2 \neq w_3$) are employed to extract fault-related features at multiple scales, and the fused results enhance shared representations while preserving scale-specific characteristics for accurate and stable fault identification. The above process can be expressed as:

$$X_{\text{init}} = \text{InitConv}\,(X) \tag{11}$$

$$X_{w_i}^{(l)} = f\left(W_{w_i}^{(l)} * X_{\text{input}}^{(l)} + b_{w_i}^{(l)}\right) \tag{12}$$

$$h_f^{(l)} = X_{w=1}^{(l)} \oplus X_{w=3}^{(l)} \oplus X_{w=5}^{(l)} \tag{13}$$

where X denotes the initial 1-D input sequence. X_{init} represents the output after the initial convolution. $X_{w_i}^{(l)}$ denotes the output of a convolutional kernel with size w_i in the lth multiscale convolutional block, in which $l \in \{1, 2, 3\}$. For

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

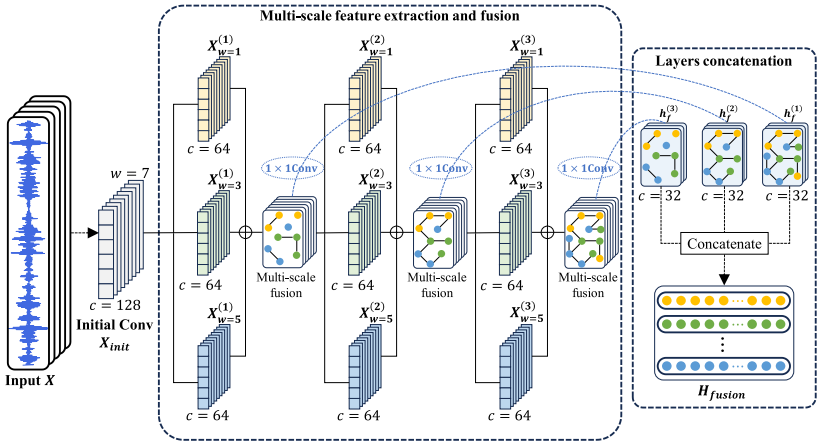


Fig. 6. Architecture of the MSF extraction and fusion module.

$l = 1$, the input is X_{init}, while for $l > 1$, it is the output of the previous multiscale fusion block. $f(\cdot)$ denotes the activation function. $W_{w_i}^{(l)}$ refers to the convolutional kernel with size w_i in the lth block, producing 64 output channels. $b_{w_i}^{(l)}$ is its corresponding bias. $h_f^{(l)}$ represents the output after scale-wise fusion in the lth block. The symbol $\oplus$ denotes the element-wise addition.

To further enhance feature representation, a layer concatenation module is introduced to fuse information across hierarchical levels. Features at different depths capture progressively abstract representations, and their integration improves the characterization of subtle engine condition changes. As shown in Fig. 6, outputs from multiple multiscale extraction blocks are first reduced in dimension and then concatenated to form a fused feature representation. The fusion process is defined as follows:

$$H_{\text{fusion}} = \text{Concat}\left(h_f^{(1)}, h_f^{(2)}, h_f^{(3)}\right) \tag{14}$$

where H_{fusion} is the final output feature map after cross-level concatenation, with a channel size of $32 \times 3 = 96$. Concat$(\cdot)$ denotes the horizontal concatenation operation.

2) Attention-Based Feature Enhancement Module: Although the MSF serves as the backbone network and effectively captures temporal features, its output typically involves a straightforward aggregation of information across the temporal and channel dimensions. This uniform treatment may lead to suboptimal performance [37]. In fault diagnosis tasks, sensor parameters contribute differently to model decisions over time, and information from certain channels may be more critical for identifying specific fault patterns. To address this, a dual-attention feature enhancement module is introduced to refine the MSF outputs by highlighting informative channels and enhancing the representation of MSFs. It strengthens relevant feature responses while suppressing interference from irrelevant ones and consists of an ICAM and an SE module based on Sliding-SE. The specific design will be introduced in detail as follows.

When handling multivariate time-series data, the traditional CAM mainly relies on global pooling statistics to generate channel weights. Although it can capture some interchannel dependencies, its limited feature fusion capability hinders the extraction of deep interaction features, reducing its effectiveness in complex pattern recognition tasks. To overcome this limitation, this study proposes an ICAM, which enhances interchannel interactions by explicitly modeling dependencies using multiscale convolution operations. As shown in Fig. 7, the input feature map $F \in \mathbb{R}^{C\times T}$ is processed by three parallel 1-D convolution layers with kernel sizes of 1, 3, and 5 to capture local features under different receptive fields. The resulting outputs at each scale are denoted as $X_1, X_2, X_3 \in \mathbb{R}^{C\times T}$. The similarity between X_2 and X_3 is then calculated to generate the channel-attention matrix $M \in \mathbb{R}^{C\times C}$, where the similarity score matrix is defined as

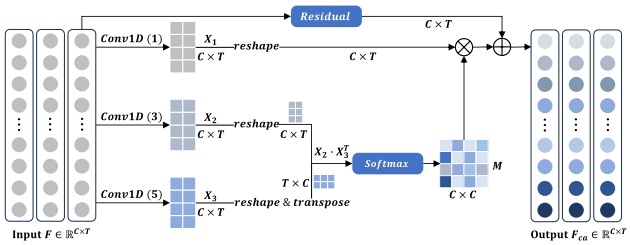


Fig. 7. Structure of the ICAM.

$$S = X_2 \cdot X_3^T \in \mathbb{R}^{C\times C}. \tag{15}$$

Subsequently, a softmax operation is applied to each row to normalize the attention weights along the channel dimension, resulting in the final attention matrix M

$$M_{i,j} = \frac{\exp\left(S_{i,j}\right)}{\sum_{k=1}^{C} \exp\left(S_{i,k}\right)} \tag{16}$$

where $M_{i,j}$ denotes the attention weight from the ith channel to the jth channel, with $i, j \in \{1, 2, \ldots, C\}$. The attention weights for each channel sum to 1. The attention matrix M is then multiplied by the feature map X_1, extracted at the first scale, to obtain attention-enhanced features that capture interchannel dependencies. These features are added to the residual features to generate the final module output, which is defined as follows:

$$F_{\text{ca}} = \alpha \cdot (M \cdot X_1) + F \tag{17}$$

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

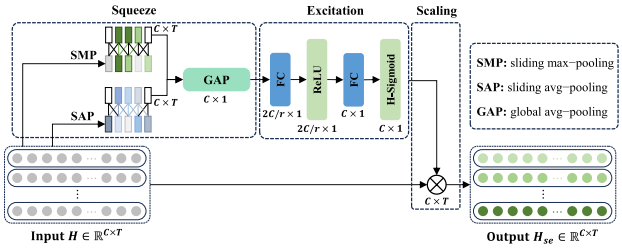


Fig. 8. Structure of the improved SE module based on Sliding-SE.

where F_{ca} represents the output of the ICAM. α is a learnable scaling factor used to balance the contribution of attention-enhanced features and residual information.

As a local enhancement unit within the feature extraction network, ICAM incorporates multiscale convolutional kernels to increase the diversity of channel feature representations, thereby emphasizing critical channel features in the early stages of extraction.

The SE module has been widely adopted in deep learning architectures for its simplicity and strong performance. However, its global average pooling (GAP) operation, while effective in static domains such as image analysis, is less suitable for fault diagnosis in marine machinery. Sensor data from such equipment often exhibit stage-wise and nonstationary patterns, where direct GAP aggregation can obscure critical local fluctuations and weaken the extraction of degradation-related features. To address this, a modified SE module based on Sliding-SE is proposed. It applies sliding max pooling (SMP) and sliding average pooling (SAP) for fine-grained feature compression, which are then integrated with GAP to generate a fused representation that preserves global trends while emphasizing key local features.

The structure of the Sliding-SE is illustrated in Fig. 8. Given an input feature map $H \in \mathbb{R}^{C\times T}$, SMP and SAP are independently applied to each channel using a pooling window of size k and stride s, respectively, producing two compressed feature maps:

$$H_{\text{smp}}(c,i) = \max_{t\in\mathcal{W}_i} H(c,t) \tag{18}$$

$$H_{\text{sap}}(c,i) = \frac{1}{k}\sum_{t\in\mathcal{W}_i} H(c,t) \tag{19}$$

where $c \in \{1,2,\ldots,C\}$, $t \in \{1,2,\ldots,T\}$, and $\mathcal{W}_i$ denotes the ith window along the temporal dimension. Subsequently, GAP is applied to the outputs of SMP and SAP to further compress them into channel-level descriptors

$$z_{\text{smp}}(c) = \frac{1}{L}\sum_{i=1}^{L} H_{\text{smp}}(c,i) \tag{20}$$

$$z_{\text{sap}}(c) = \frac{1}{L}\sum_{i=1}^{L} H_{\text{sap}}(c,i) \tag{21}$$

where L represents the number of temporal steps after pooling, determined by the input length T, window size k, and stride s. Finally, the two descriptors are concatenated to form a fused representation

$$z = \left[z_{\text{smp}}, z_{\text{sap}}\right] \in \mathbb{R}^{2C}. \tag{22}$$

The fused vector z is subsequently fed into two FC layers to perform a nonlinear transformation, yielding the attention weights for each channel

$$s = \sigma\left(W_2 \cdot \text{ReLU}\left(W_1 \cdot z\right)\right) \in \mathbb{R}^{C} \tag{23}$$

where $W_1 \in \mathbb{R}^{(2C/r)\times 2C}$ and $W_2 \in \mathbb{R}^{C\times(2C/r)}$, in which r denotes the channel reduction ratio. $\sigma(\cdot)$ represents the sigmoid activation function. The resulting attention vector s is reshaped to match the dimensions of the input and is applied to the feature map H via channel-wise multiplication for recalibration. The final output of the module is defined as

$$H_{\text{se}}(c,t) = s(c) \cdot H(c,t). \tag{24}$$

The Sliding-SE improves the sensitivity of each channel to salient local features through sliding pooling operations, thereby enhancing the retention of critical information during the fusion process.

3) Overall Process of the AMSF: The overall process of AMSF is illustrated in Fig. 9 and includes feature extraction, multiscale fusion with attention enhancement, and final output prediction. First, the input multivariate time-series data pass through three stacked multiscale convolution modules. Each module applies multiple 1-D convolutional kernels of different sizes to capture local patterns across varying receptive fields. An ICAM is inserted after the first multiscale convolution module to enhance the response of key feature channels, thereby improving the perception of shallow features toward potential fault patterns. Then, the fused features from the three convolutional modules are reduced in dimension and concatenated along the channel axis to achieve unified fusion across hierarchical levels. The Sliding-SE module is applied to the fused feature maps to further improve the network capacity to capture local abnormal features while preserving global trend information. Finally, the enhanced feature representations are passed through an FC layer to complete feature extraction.

AMSF progressively enhances feature discriminability and temporal modeling through the joint effect of local convolution modeling and channel-attention mechanisms, providing richer and more distinctive representations for subsequent multimodal fusion.

D. Low-Rank Multimodal Fusion

PBM and DDM reflect the condition of the marine engine from different perspectives and serve as two distinct forms of engine information representation. Multimodal feature fusion enables the capture of correlations and complementary information between these representations. Compared with traditional fusion strategies such as simple concatenation or full bilinear pooling, LMF achieves a better balance between representation capability and computational efficiency, making it well suited for marine engine condition monitoring. Therefore, the LMF method is introduced to integrate the outputs of both the PBM and the DDM. While capturing high-order interactions between modalities, LMF

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

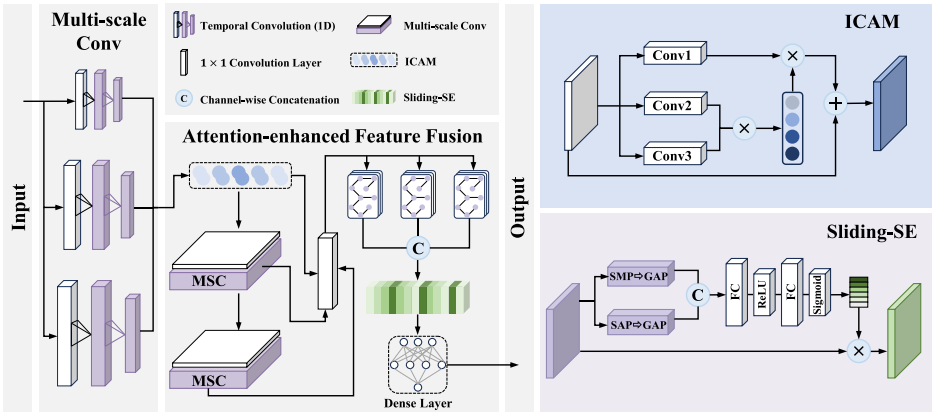


Fig. 9. Process of the AMSF.

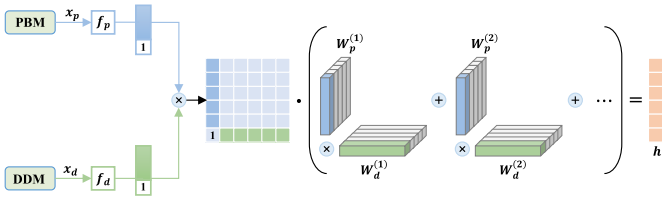


Fig. 10. Structure of dual-modal low-rank tensor fusion.

employs the low-rank decomposition to reduce computational complexity [38], thereby improving fusion efficiency and enhancing feature representation in the DT modeling process.

The LMF fusion process is illustrated in Fig. 10. First, the outputs from PBM and DDM are denoted as $x_p \in \mathbb{R}^{d_p}$ and $x_d \in \mathbb{R}^{d_d}$, respectively. These outputs are passed through an MSF extraction network to obtain unified feature representations, f_p and f_d. To construct a bimodal tensor, a bias term (constant 1) is appended to each feature vector, and an outer product is computed to generate the multimodal tensor representation Z

$$Z = [f_p; 1] \otimes [f_d; 1] \tag{25}$$

where $\otimes$ denotes the tensor outer product operation, resulting in $Z \in \mathbb{R}^{(d_p+1)\times(d_d+1)}$. To reduce the computational cost associated with tensor operations, a low-rank decomposition strategy is adopted. The high-dimensional weight tensor W is represented as the sum of rank-r outer products of vectors

$$W = \sum_{i=1}^{r} w_p^{(i)} \otimes w_d^{(i)} \tag{26}$$

where $w_p^{(i)} \in \mathbb{R}^{d_p+1}$ and $w_d^{(i)} \in \mathbb{R}^{d_d+1}$ are the low-rank factors corresponding to the PBM and DDM modalities, respectively. The final fused representation $h \in \mathbb{R}^{d_h}$ is obtained through a weighted tensor summation

$$h = \sum_{i=1}^{r} \left(w_p^{(i)} \cdot [f_p; 1]\right) \cdot \left(w_d^{(i)} \cdot [f_d; 1]\right). \tag{27}$$

By constructing and aggregating low-rank vector pairs, LMF not only compresses intramodal information but also preserves intermodal interactions. The resulting fused output h serves as the input to the fault diagnosis model, enabling accurate identification of key performance-degradation patterns.

TABLE II
PERFORMANCE PARAMETERS OF ENGINE COMPONENTS

Parameter	Definition	Parameter	Definition
N_e	Main engine power	P_m	Peak CYL pressure
P_{bme}	Brake mean effective pressure	T_{ae}	Average exhaust gas temperature
N_{tc}	TC speed	T_{cac}	CAC outlet temperature
T_{to}	Turbo outlet temperature	P_{cac}	CAC outlet air pressure

IV. EXPERIMENTS

A. Data Generation and Preprocessing

1) Data Generation: A dataset containing both healthy and fault-labeled conditions was collected to validate the proposed fault diagnosis method. The dataset comprises local parameters, such as the temperatures and pressures of key components, and global measurements that reflect overall system performance, as shown in Table II. However, for essential onboard machinery such as marine engines and generators, it is almost impossible to obtain sufficient fault data under real operating conditions, limiting the feasibility of comprehensive diagnostic experiments [39]. To address this limitation, the constructed PBM + PDC model was employed to generate supplementary fault data while carefully preventing potential simulator leakage during data partitioning.

In addition, the generated fault data should be generalizable to real operating conditions. Notably, when a fault does not alter the fundamental in-CYL combustion mechanism, the combustion, heat transfer, and other submodels developed under healthy conditions remain applicable [9], allowing the PBM + PDC to accurately reproduce the operating characteristics of different fault categories. Furthermore, the simulated and measured data demonstrate an approximately consistent mapping between engine status and operating signals, ensuring that the overall results under multiple operating conditions

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

TABLE III
MEASUREMENT PARAMETERS AND CORRESPONDING NOISE TYPES

Types	Parameters	WGN	HCE	IBN	SDN	Total range
Speed & Power	N_e , N_{tc}	✓	✓	✓		±0.25%
Temperature	T_{to} , T_{ae} , T_{cac}	✓	✓	✓		±0.3%
Pressure	P_{bme} , P_a , P_{cac}	✓	✓	✓	✓	±0.5%

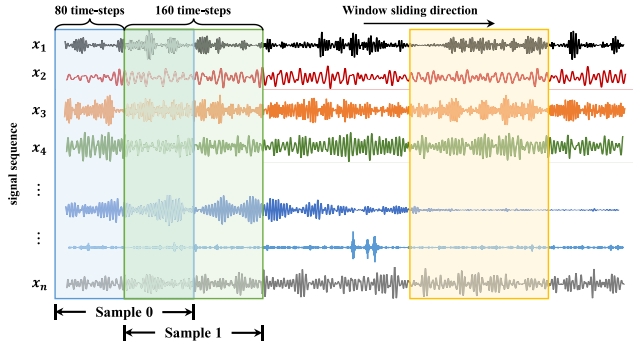


Fig. 11. Illustration of the data segmentation process.

reliably reflect the fault diagnosis performance of the proposed method.

2) Data Preprocessing: To better simulate the operating conditions of the marine engine, this study incorporates the uncertainties associated with sensor measurements. Based on the relevant measurement principles, the simulated noise added to the engine signals includes white Gaussian noise (WGN), inherent human calibration error (HCE), intrinsic bias noise (IBN), and sinusoidal disturbance noise (SDN) [40]. It should be noted, however, that evaluating the influence of these stochastic noise types on model accuracy is not the primary focus of this study. The amplitudes of each noise type were randomly and uniformly generated within the predefined ranges given in Table III, corresponding to ±0.25%, ±0.3%, and ±0.5% of the nominal values for speed/power, temperature, and pressure measurements, respectively.

To reduce the computational burden of subsequent modeling, the collected raw data were normalized using the min–max method, defined as follows:

$$x_i' = \frac{x_i - x_{i,\min}}{x_{i,\max} - x_{i,\min}} \tag{28}$$

where x_i and x_i' denote the original and normalized values, respectively. $x_{i,\max}$ and $x_{i,\min}$ represent the maximum and minimum values of the dataset, respectively.

After normalization, the time-series data were segmented using a sliding-window method, as illustrated in Fig. 11. The window length was set to 160 time steps, with a stride of 80. With the PBM simulation time step set to 0.1 s, each resulting sample contains a sequence length of 160 and a feature dimension of 8. This configuration ensures that each segmented sample effectively captures the dynamic state variations of the marine engine across multiple power cycles.

3) Datasets: The experimental data consist of both sensor measurements and simulation outputs processed with appropriate noise levels. Each fault category comprises 5000 samples.

TABLE IV
DIVISION OF DATASETS

Data set	Proportion	Number of samples per class	Role
Training set	70%	3500	Train the deep learning model
Validation set	20%	1000	Tune hyperparameters and assess generalization during training
Testing set	10%	500	Evaluate final diagnostic accuracy and generalization

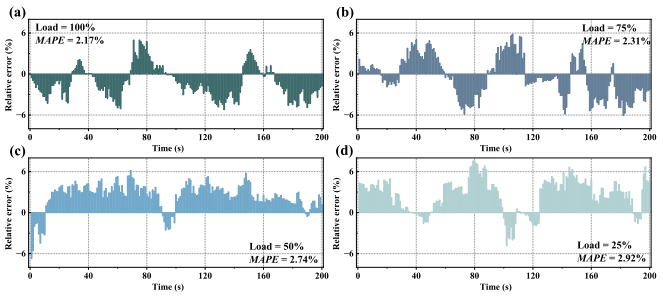


Fig. 12. Relative errors between simulated and measured average exhaust gas temperatures under different engine load conditions. (a) 100% load, (b) 75% load, (c) 50% load, and (d) 25% load.

The datasets were partitioned according to Table IV, with 70% of the samples for training, 20% for validation, and 10% for testing. Four representative datasets corresponding to engine loads of 100%, 75%, 50%, and 25% were constructed to evaluate the performance of the proposed method. Each dataset maintains the same fault category distribution and noise level configuration.

B. Validation of Adaptive PDC

This section aims to validate the effectiveness of the proposed PDC module in improving the accuracy of the PBM. To verify the reliability of the model across various operating conditions, the simulated average exhaust gas temperatures were compared with the corresponding measured values under different load conditions (100%, 75%, 50%, and 25%). The mean absolute percentage error (MAPE) was used as the evaluation metric, defined as follows:

$$\text{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{X_{i,\text{sim}} - X_{i,\text{obs}}}{X_{i,\text{obs}}}\right| \times 100\% \tag{29}$$

where N denotes the number of samples under each load condition. $X_{i,\text{sim}}$ and $X_{i,\text{obs}}$ represent the simulated and observed average exhaust gas temperatures, respectively.

Fig. 12 shows the relative error distribution between PBM-simulated and measured average exhaust gas temperatures under four load conditions. The results reveal noticeable discrepancies between PBM outputs and actual engine measurements, with the relative error increasing as the load decreases. The maximum relative error observed is 7.6%. Moreover, under medium- and low-load conditions, the simulated values are consistently higher than the measured ones, indicating a clear pattern of systematic error.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

TABLE V
HI DEFINITIONS AND ASSOCIATED SENSOR PARAMETERS FOR EACH COMPONENT

Component	HI index	Sensor parameter used	Parameter unit
Turbocharger	TC1	N_{tc}	rpm
	TC2	T_{to}	°C
Cylinder	CYL1	P_m	bar
	CYL2	T_{ae}	°C
Charge air cooler	CAC1	T_{cac}	°C
	CAC2	P_{cac}	bar

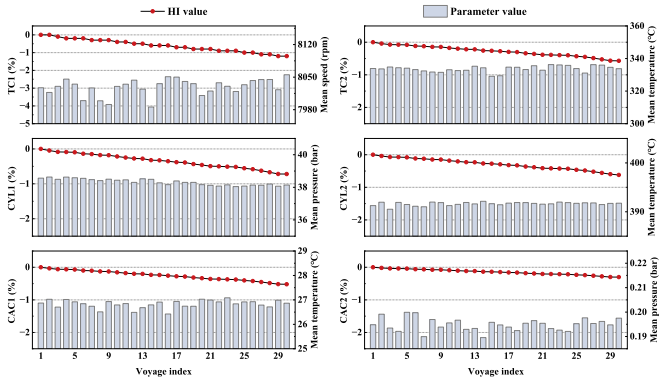


Fig. 13. Component-level HI variation predicted by the PDC module across different voyages.

The PDC module is used to adaptively correct the long-term predictive accuracy of the PBM. In this study, HIs were developed for three key components of the PBM—TC, CYL, and CAC—to characterize performance-degradation trends. Table V lists the selected components, HI definitions, and associated measurement parameters used in their construction. The HIs predicted by the PDC module form the basis for iterative simulation updates, ensuring ongoing consistency between the PBM and the actual mechanical condition. To ensure accurate HI construction, measurement data from 30 consecutive voyages under stable engine load conditions were selected. Data were sampled at 1-s intervals, yielding 5400 HI training samples per component.

Fig. 13 presents the trends of component-level HIs predicted by the PDC module across consecutive voyages. The mean values of sensor measurements exhibit random fluctuations due to noise interference, which hampers their ability to accurately reflect the actual condition of the components. In contrast, the HI values predicted using KL divergence—derived from differences in probability distributions—as input parameters exhibit a general downward trend, effectively capturing the long-term performance degradation of engine components.

To further assess the effectiveness of the PDC module, adaptive correction was applied to relevant modeling parameters in the PBM based on the predicted HIs. Table VI summarizes the relative errors between the simulation outputs after PDC correction and the corresponding measurements, covering representative global performance and thermodynamic parameters. As shown in the table, the relative errors of the average values under all load conditions remain below 4%, indicating that the corrected PBM can reliably simulate

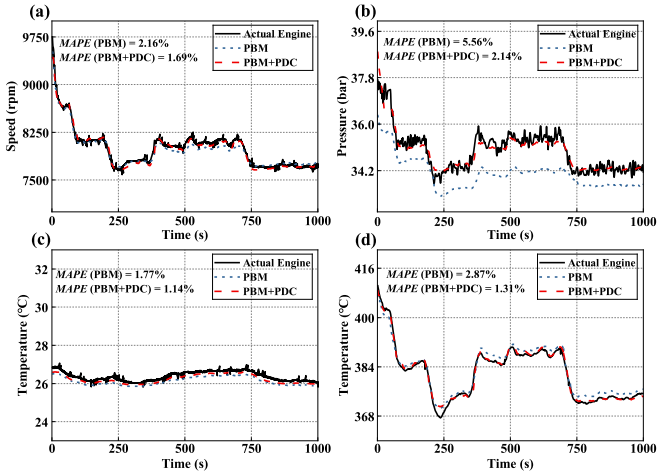


Fig. 14. Comparison of simulated and measured parameters before and after PDC correction under low-load conditions. (a) TC speed (N_{tc}). (b) Peak CYL pressure (P_m). (c) CAC outlet temperature (T_{cac}). (d) Average exhaust gas temperature (T_{ae}).

the real engine behavior. In addition, Fig. 14 illustrates a comparison of model outputs before and after correction under low-load conditions. The results clearly show that the simulation errors of the PBM corrected by the PDC module are lower than those of the original PBM. In particular, the relative error of P_m decreased from 5.56% to 2.14% after correction [Fig. 14(b)]. Moreover, the MAPE values of N_{tc}, T_{cac}, and T_{ae} were reduced by 21.76%, 35.59%, and 54.36%, respectively, when using the PBM + PDC model compared to the original PBM. These findings demonstrate that the PDC module effectively mitigates the systematic errors caused by state variations during prolonged operation of marine engine components. Therefore, the PBM + PDC model enables reasonably accurate condition monitoring within the proposed DT framework.

C. Fault Injection

The PBM + PDC-based marine engine model can replicate the behavior of real engines under various fault conditions. The fault categories considered in this study primarily focus on gradual time-dependent degradation, including wear, fouling, and erosion, which ultimately manifest as changes in the thermodynamic parameters of the engine [41]. Among key engine components, the TC, CYL, and CAC are particularly prone to early stage faults that are difficult to detect in real time [42]. Therefore, both single and combined fault scenarios are introduced by modifying the efficiency and flow characteristics of these three components.

Table VII presents the fault injection scheme, which includes reductions in TC efficiency (TCE), air cooler efficiency (ACE), and anomalies in exhaust valve gap (EVG). The values 0.77, 0.89, and 0.80 correspond to the normal operating conditions of the actual engine for TCE, ACE, and EVG, respectively, and are considered representative of the healthy state. Fault-TCE-i, Fault-ACE-i, and Fault-EVG-i ($i = 1, 2$) denote varying levels of faults in the TC, CAC, and the exhaust valve of the CYL, respectively, while Fault-TCE-ACE and Fault-TCE-EVG represent combined faults involving

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF SIMULATED AND MEASURED PARAMETERS UNDER DIFFERENT LOAD CONDITIONS

Engine load (%)		P_{bme} (bar)	Power (kW)	TC		CYL		CAC	
				N_{tc} (rpm)	T_{to} (°C)	P_m (bar)	T_{ae} (°C)	T_{cac} (°C)	P_{cac} (bar)
100	Measured	19.96	4050.00	21300	412	112.90	531	46	1.84
	Simulated	20.15	4059.42	21556	408.62	114.37	530.54	46.70	1.87
	Error (%)	0.95	0.23	1.20	-0.82	1.30	-0.09	1.52	1.63
75	Measured	16.85	3038.00	17770	402	100.34	528	42	1.73
	Simulated	17.44	3080.53	18053	396.73	98.63	531.22	42.51	1.77
	Error (%)	3.50	1.40	1.59	-1.31	-1.70	0.61	1.21	2.31
50	Measured	12.13	2025.00	15070	385	84.10	501	38	1.19
	Simulated	12.42	2066.53	15326	390.62	82.43	505.50	37.63	1.23
	Error (%)	2.39	2.05	1.70	1.46	-2.14	0.90	-0.97	3.36
25	Measured	5.57	1013.00	10590	343	50.73	412	29	0.58
	Simulated	5.75	1042.07	10420	352.23	49.69	416.70	28.70	0.60
	Error (%)	3.23	2.87	-1.61	2.69	-2.05	1.14	-1.03	3.45

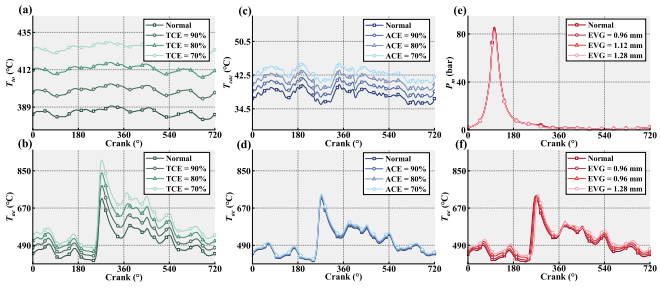


Fig. 15. Impact of different fault categories on engine performance parameters. (a) Effect of TCE on T_{to}. (b) Effect of TCE on T_{ae}. (c) Effect of ACE on T_{cac}. (d) Effect of ACE on T_{ae}. (e) Effect of EVG on P_m. (f) Effect of EVG on T_{ae}.

multiple components. It should be noted that, in accordance with the real-world behavior of marine engines, the severity of each fault is randomly selected within a predefined range. Furthermore, as this study focuses on DT-based fault diagnosis methods for marine machinery, sensor faults are not considered within its scope.

After determining the fault categories to be simulated, the state signals listed in Table II were selected as feature parameters for subsequent fault diagnosis. A dataset containing nine labels was then generated by combining sensor measurements with the fault injection scheme. Fig. 15 illustrates the variations in engine parameters during a single power cycle under different fault conditions. It is evident that distinct fault categories lead to different fluctuations in the thermodynamic parameters inside the engine. Moreover, as the fault severity increases, the deviations in parameters become more pronounced.

D. Fault Diagnosis Based on the Proposed DT Approach

The case study in Section IV-B validated the effectiveness of the proposed PDC module in enhancing the accuracy of the PBM. This improvement enables more accurate prediction of performance parameters for marine engine components, making the model suitable for condition monitoring of marine machinery. In this section, the fault injection scheme described in Section IV-C is applied to generate performance parameters of each engine component within the degradation range. These data are used to construct datasets under different engine loads for validating the fault diagnosis functionality. The dataset partitioning details are provided in Table IV.

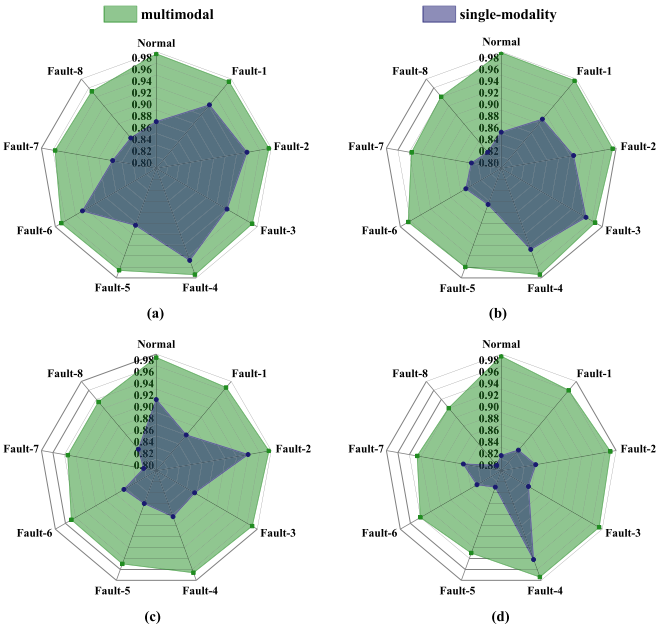


Fig. 16. Comparison of diagnostic accuracy between single-modality and multimodal methods under different load conditions. (a)–(d) Samples 1–4, respectively.

1) Comparison of Multimodal and Single-Modality Methods:

To evaluate the effectiveness of the proposed multimodal fusion-based DT approach, the AMSF model—representing an advanced deep learning method—is used as a baseline. The AMSF model is trained solely on the output data from the PBM (i.e., single modality). The fault diagnosis results for the nine predefined labels are presented in Fig. 16. Under all four engine load conditions, the multimodal fusion-based method consistently demonstrates superior diagnostic performance. Its accuracy across all normal and fault categories is significantly higher than that of the single-modality model. In particular, the AMSF model exhibits a noticeable decline in detection accuracy for the normal and combined fault categories. These results highlight the advantages of multimodal fusion over single-modality methods. Moreover, the diagnostic accuracy of the multimodal method shows minimal variation across different engine loads.

Table VIII summarizes the average accuracies of the two models on the four sample datasets. Compared with the

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
FAULT CATEGORIES CORRESPONDING TO TCE, ACE, AND EVGS

Fault item	TCE (%)	ACE (%)	EVG (mm)	Label
Normal	0.77*90% < TCE ≤ 0.77*100%	0.89*90% < ACE ≤ 0.89*100%	0.80*100% ≤ EVG < 0.80*120%	0
Fault-TCE-1	0.77*80% < TCE ≤ 0.77*90%	0.89*90% < ACE ≤ 0.89*100%	0.80*100% ≤ EVG < 0.80*120%	1
Fault-TCE-2	0.77*70% < TCE ≤ 0.77*80%	0.89*90% < ACE ≤ 0.89*100%	0.80*100% ≤ EVG < 0.80*120%	2
Fault-ACE-1	0.77*90% < TCE ≤ 0.77*100%	0.89*80% < ACE ≤ 0.89*90%	0.80*100% ≤ EVG < 0.80*120%	3
Fault-ACE-2	0.77*90% < TCE ≤ 0.77*100%	0.89*70% < ACE ≤ 0.89*80%	0.80*100% ≤ EVG < 0.80*120%	4
Fault-EVG-1	0.77*90% < TCE ≤ 0.77*100%	0.89*90% < ACE ≤ 0.89*100%	0.80*120% ≤ EVG < 0.80*140%	5
Fault-EVG-2	0.77*90% < TCE ≤ 0.77*100%	0.89*90% < ACE ≤ 0.89*100%	0.80*140% ≤ EVG < 0.80*160%	6
Fault-TCE-ACE	0.77*80% < TCE ≤ 0.77*90%	0.89*80% < ACE ≤ 0.89*90%	0.80*100% ≤ EVG < 0.80*120%	7
Fault-TCE-EVG	0.77*80% < TCE ≤ 0.77*90%	0.89*90% < ACE ≤ 0.89*100%	0.80*120% ≤ EVG < 0.80*140%	8

TABLE VIII
AVERAGE ACCURACIES OF SINGLE-MODALITY AND MULTIMODAL METHODS

Sample	Engine load	Accuracy (%)		Improvement of accuracy (%)
		single-modality	multimodal	
1	100%	92.02	98.80	↑ 6.78
2	75%	89.31	98.31	↑ 9.00
3	50%	87.96	97.76	↑ 9.80
4	25%	85.60	97.22	↑ 11.62

single-modality model, the multimodal fusion method improves fault diagnosis accuracy under the various load conditions by 6.78%–11.62%, with an average gain of 9.30%. Overall, the proposed DT approach significantly improves diagnostic reliability, particularly under challenging operating conditions such as low-load scenarios.

Building on the above analysis, the contribution of the proposed LMF strategy to the diagnostic performance of the DT approach was further assessed. Under identical experimental conditions, the LMF method was compared with the conventional concatenation approach and the full bilinear pooling method. As shown in Fig. 17, LMF consistently outperformed concatenation across all sample datasets under the four load conditions. Notably, although the diagnostic accuracy of bilinear pooling was comparable to that of LMF in certain datasets, its average inference time was longer, whereas LMF maintained computational efficiency close to that of concatenation. These results highlight that LMF achieves a balanced tradeoff between diagnostic accuracy and inference efficiency within the proposed fusion framework.

2) Comparison With Recently Published Fault Diagnosis Methods: The confusion matrices of the proposed DMIF-DT fault diagnosis method are presented in Fig. 18. As illustrated,

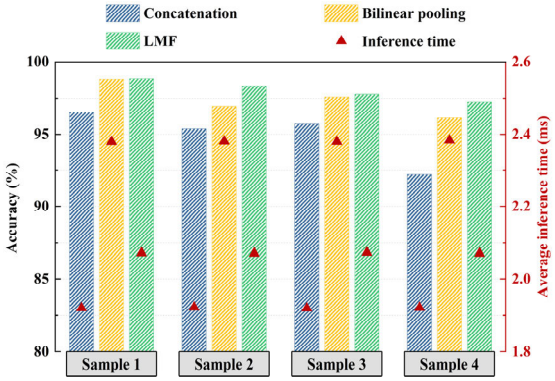


Fig. 17. Performance comparison of fusion strategies under different load conditions.

the approach achieves very high diagnostic accuracy. At the highest load condition [Fig. 18(a)], almost all samples in each fault category are correctly identified, with only a few misclassifications. Similarly, a strong diagnostic performance is observed under the other load conditions [Fig. 18(b)–(d)], demonstrating the excellent generalization capability of the proposed method. However, relatively more misclassification cases occur in the combined fault scenarios (labels 7 and 8). This is because discriminative feature extraction becomes more challenging when multiple faults occur simultaneously.

Table IX summarizes the diagnostic accuracies for each fault category under various load conditions. As shown, the prediction accuracy for every label is no lower than 94.00%. Even when the load decreases (e.g., to 50% and 25%), making the diagnosis more challenging, the overall diagnostic performance remains stable, with an overall accuracy of 97.22% at the lowest load condition. These results confirm the robustness of the proposed DMIF-DT method in capturing fault features; its advantage is particularly evident under low-load conditions, where signal disturbance and noise are more pronounced.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
FAULT DIAGNOSIS ACCURACIES OF THE PROPOSED DMIF-DT METHOD ACROSS FAULT CATEGORIES

Sample type	Engine load	Accuracy for different fault categories (%)									
		0	1	2	3	4	5	6	7	8	Overall
Sample 1	100%	99.6	99.4	99.6	99.0	99.4	98.6	98.8	97.6	97.2	98.80
Sample 2	75%	99.8	99.6	99.4	98.6	99.4	98.0	98.4	95.6	96.0	98.31
Sample 3	50%	99.4	98.6	99.6	99.0	98.6	97.0	96.8	95.4	95.4	97.76
Sample 4	25%	99.6	98.0	99.0	99.4	99.4	95.0	96.0	94.6	94.0	97.22

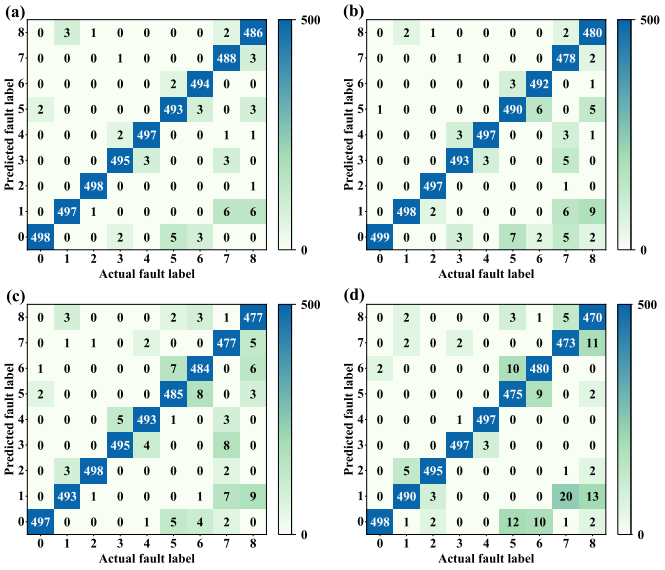


Fig. 18. Confusion matrices for fault diagnosis with the proposed DMIF-DT method under different load conditions. (a) 100% load, (b) 75% load, (c) 50% load, and (d) 25% load.

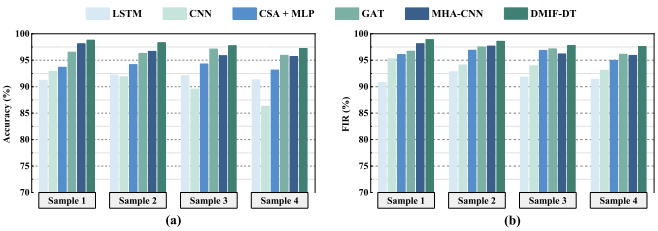


Fig. 19. Performance comparison of the proposed DMIF-DT and recent fault diagnosis methods. (a) Accuracy. (b) FIR.

To further verify and evaluate the diagnostic performance of the proposed DMIF-DT approach, several additional deep learning models were selected for comparative experiments. Long short-term memory (LSTM) networks and CNNs were included as representative AI methods from different model families [15]; both have been widely adopted in engineering practice and demonstrate excellent performance. In recent studies on marine machinery fault diagnosis, the CSA + MLP [43], GAT [44], and MHA-CNN [39] models have emerged as state-of-the-art techniques and were therefore used as additional baselines. Model performance was compared using overall accuracy and the fault-isolation rate (FIR), defined as the proportion of correctly identified fault types among all fault samples. Detailed results are presented in Fig. 19 and Table X.

As shown by the overall accuracies in Fig. 19(a), the DMIF-DT approach exhibits clear advantages across all four load conditions, achieving higher diagnostic accuracy than any of the other methods. Under the highest load condition (100%), DMIF-DT attains an accuracy of 98.80%, outperforming the next-best MHA-CNN model (98.11%), whereas the remaining baselines drop to as low as 91.20% (LSTM). All models experience a decline in performance under low-load conditions, with the CNN model showing the most notable drop—falling to 86.29% at 25% load. In contrast, DMIF-DT maintains high diagnostic reliability in these challenging scenarios. The better performance can be explained as follows. Because all features are learned end-to-end, DMIF-DT employs two independent processing streams to extract modality-specific features and fuses them at a higher level, thereby reducing the influence of redundant information and noise.

The FIR measures a model's ability to distinguish accurately among different fault categories. As shown in Fig. 19(b), the FIR of the DMIF-DT approach remains consistently above 97% under all load conditions and reaches a maximum of 98.90%. In contrast, traditional methods such as LSTM and CNN exhibit much greater variation in FIR, making it difficult for them to provide stable and reliable fault discrimination across different operating conditions. Although attention-based models such as GAT and MHA-CNN outperform the conventional baselines in terms of FIR, they still fall slightly short of the performance achieved by the proposed method.

E. Ablation Study

To explore the contribution of each module in the proposed DMIF-DT method to the fault diagnosis task, a series of ablation experiments was conducted. Specific modules in DMIF-DT were removed or replaced sequentially, resulting in seven ablation models: w/o MSF, where a single-scale CNN replaces the MSF module; w/o ICAM, where the ICAM is removed; w/o Sliding-SE, where the original SE module is used instead of the Sliding-SE; w/o LMF, where the LMF module is replaced by simple concatenation; w/o PDC, where the PDC module is removed; w/o Att, where both attention modules (ICAM and Sliding-SE) are removed; and w/o MSF + LMF, where the MSF and LMF modules are jointly removed. These variants were used to evaluate the impact of each module individually and in combination. Fig. 20 details the changes in accuracy and FIR for every ablation model under different load conditions, while Table XI summarizes the average performance metrics of each model.

As shown in Table XI, each ablation model exhibits a certain degree of performance decline compared with the complete DMIF-DT model. Among the single-module variants, the w/o PDC model shows the most significant decrease, with accuracy and FIR dropping by 4.37% and 4.03%, respectively.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

TABLE X
COMPARISON OF DIAGNOSTIC PERFORMANCE AMONG DIFFERENT METHODS

Method	Sample 1		Sample 2		Sample 3		Sample 4	
	Accuracy (%)	FIR (%)	Accuracy (%)	FIR (%)	Accuracy (%)	FIR (%)	Accuracy (%)	FIR (%)
LSTM	91.20	90.80	92.22	92.81	92.09	91.81	91.31	91.40
CNN	92.89	95.26	91.87	94.10	89.53	93.95	86.29	93.11
CSA + MLP	93.67	96.07	94.18	96.89	94.29	96.85	93.13	94.98
GAT	96.51	96.70	96.29	97.50	97.11	97.14	95.93	96.13
MHA-CNN	98.11	98.13	96.67	97.67	95.87	96.19	95.71	95.90
DMIF-DT	98.80	98.90	98.31	98.58	97.76	97.77	97.22	97.59

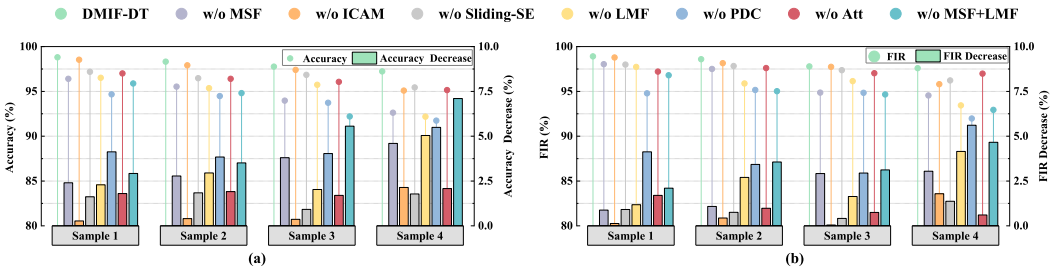


Fig. 20. Diagnostic performance comparison of different ablation models under different load conditions. (a) Accuracy and accuracy decrease. (b) FIR and FIR decrease.

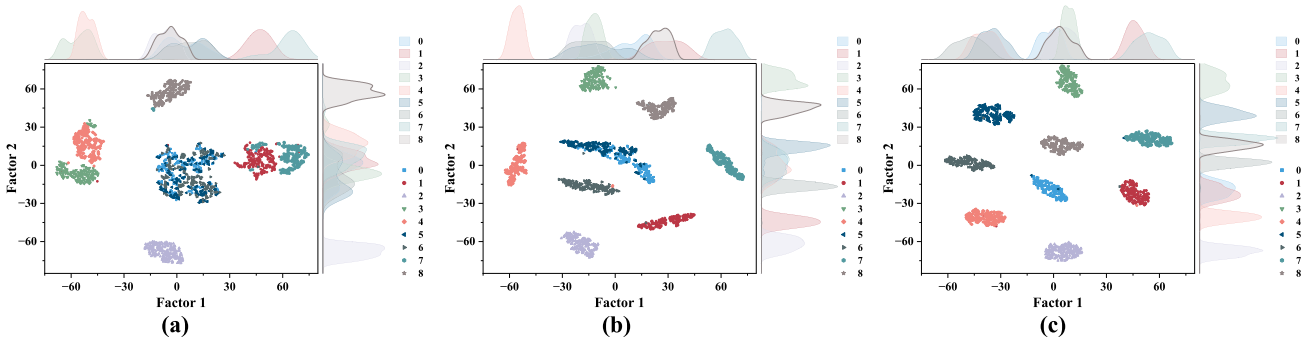


Fig. 21. Visualization of intermediate features at three key stages using t-SNE. (a) Before attention stage (MSC output). (b) After attention stage (AMSF output). (c) LMF.

TABLE XI
AVERAGE DIAGNOSTIC ACCURACY AND FIR COMPARISON OF ABLATION MODELS

Method	Accuracy (%)	Accuracy Decrease (%)	FIR (%)	FIR Decrease (%)
DMIF-DT	98.02	/	98.21	/
w/o MSF	94.63	↓ 3.39	96.23	↓ 1.98
w/o ICAM	97.23	↓ 0.79	97.61	↓ 0.60
w/o Sliding-SE	96.48	↓ 1.54	97.35	↓ 0.86
w/o LMF	94.95	↓ 3.07	95.80	↓ 2.41
w/o PDC	93.65	↓ 4.37	94.18	↓ 4.03
w/o Att	96.15	↓ 1.87	97.20	↓ 1.01
w/o MSF+ LMF	93.25	↓ 4.77	94.85	↓ 3.36

In addition, the combined ablation models show greater performance decline than the single-module ones.

The performance loss caused by w/o MSF is also substantial, with accuracy and FIR decreasing by 3.39% and 1.98%, respectively. This finding indicates that the MSF module—by extracting and fusing MSFs—effectively enhances feature representation and thus contributes significantly to diagnostic performance. Likewise, replacing the LMF module with simple concatenation leads to a noticeable decline in performance, with accuracy dropping by 3.07% and FIR by 2.41%, highlighting its important role in improving data reliability. By employing low-rank fusion, the DT framework can markedly reduce redundancy among cross-modal information and thereby extract more valuable features.

In comparison, removing or replacing the ICAM and Sliding-SE modules results in only modest performance degradations: accuracy declines by 0.79% and 1.54%, while FIR decreases by 0.60% and 0.86%. These findings indicate that the ICAM and Sliding-SE modules make positive contributions to the overall diagnostic performance by enhancing channel-wise feature representation and improving temporal feature capture, respectively.

To improve the transparency of the learned representations, the proposed framework visualizes the features at three key stages using t-SNE: (a) before attention (MSC output), (b) after attention (AMSF output), and (c) LMF. As shown in Fig. 21, the preattention features exhibit partially overlapping

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

clusters, indicating limited separability. After attention, the clusters become more compact and better separated, suggesting that the AMSF enhances discriminability through multiscale extraction with channel-wise recalibration. Following LMF, interclass boundaries are further enlarged, and the clusters are well separated, indicating that complementary PBM–DDM information is effectively integrated into a unified and more discriminative space.

In summary, the above results highlight the importance of each module within the digital twin framework, as using only part of the components results in reduced diagnostic performance. The complete DMIF-DT method achieves the highest accuracy and FIR, consistently delivering the best overall performance under various operating conditions.

V. Conclusion

To enable intelligent fault diagnosis for marine machinery, this study proposes a deep multimodal information fusion DT framework. Serving as two complementary modalities, the PBM and DDM describe the operating state of marine machinery from the mechanistic and data perspectives, respectively. An AMSF module performs deep extraction and enhancement of multimodal features, while a low-rank fusion strategy constructs a unified high-dimensional representation for high-accuracy fault diagnosis. To mitigate model mismatch caused by performance degradation, a PDC module leverages measured data and probability distribution analysis to build a feedback-enabled interaction between the PBM and DDM, thereby improving model adaptability. The effectiveness of the proposed framework is demonstrated through case studies, and the principal findings are as follows.

1) With PDC correction, the PBM improves the simulation accuracy of key components by 21.76%–61.51%, achieving a closer alignment with measured engine performance across different operating conditions and demonstrating potential for reliable long-term application.
2) Through deep fusion, the PBM and DDM form a dual-direction information path. Compared with the single modality, the multimodal fusion approach increases average fault diagnosis accuracy by 9.30%, confirming the effectiveness of the fusion strategy in enhancing diagnostic robustness.
3) Under the tested load conditions, DMIF-DT effectively identifies fault types and severities across multiple scenarios, demonstrating strong generalization within the experimental scope. Moreover, DMIF-DT shows a clear accuracy advantage over other comparative models.
4) Ablation studies indicate that AMSF—via the synergy of multiscale convolutional modeling and an improved channel-attention mechanism—effectively boosts diagnostic performance by reinforcing fault feature representations.

In summary, the proposed DT framework offers a valuable reference for predictive maintenance of marine machinery. Despite its high diagnostic performance, this study assumes that the PBM accurately captures the main thermodynamic behavior and that sensor data remain within valid ranges. Therefore, faults that violate these assumptions (e.g., mechanical breakage or severe sensor failure) may limit the applicability of the framework. In addition, the proposed framework has been validated under the tested load conditions, and future work will focus on extending it toward engineering deployment to demonstrate its potential for practical application in shipboard monitoring systems.

References

[1] Y. Li, Z. Guo, Z. Li, Z. Deng, and K. Noman, "Instantaneous angular speed-based fault diagnosis of multicylinder marine diesel engine using intrinsic multiscale dispersion entropy," *IEEE Sensors J.*, vol. 23, no. 9, pp. 9523–9535, May 2023, doi: 10.1109/JSEN.2023.3257832.

[2] Z. Jin, J. Liu, M. Xu, H. Miao, and Z. Song, "An enhanced digital twin-driven fault detection and isolation method based on sensor series imaging mechanism for gas turbine engine," *Appl. Thermal Eng.*, vol. 257, Dec. 2024, Art. no. 124308, doi: 10.1016/j.applthermaleng.2024.124308.

[3] Y. Qin, T. Zhang, Q. Qian, and Y. Mao, "Large model for rotating machine fault diagnosis based on a dense connection network with depthwise separable convolution," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–12, 2024, doi: 10.1109/TIM.2024.3396841.

[4] A. Coraddu, L. Oneto, F. Cipollini, M. Kalikatzarakis, G.-J. Meijn, and R. Geertsma, "Physical, data-driven and hybrid approaches to model engine exhaust gas temperatures in operational conditions," *Ships Offshore Struct.*, vol. 17, no. 6, pp. 1360–1381, Jun. 2022, doi: 10.1080/17445302.2021.1920095.

[5] Y. Raptodimos and I. Lazakis, "Application of NARX neural network for predicting marine engine performance parameters," *Ships Offshore Struct.*, vol. 15, no. 4, pp. 443–452, Apr. 2020, doi: 10.1080/17445302.2019.1661619.

[6] K. Milioulis, V. Bolbot, and G. Theotokatos, "Model-based safety analysis and design enhancement of a marine LNG fuel feeding system," *J. Mar. Sci. Eng.*, vol. 9, no. 1, p. 69, Jan. 2021, doi: 10.3390/jmse9010069.

[7] J. A. P. Rubio, F. Vera-García, J. H. Grau, J. M. Cámara, and D. A. Hernandez, "Marine diesel engine failure simulator based on thermodynamic model," *Appl. Thermal Eng.*, vol. 144, pp. 982–995, Nov. 2018, doi: 10.1016/j.applthermaleng.2018.08.096.

[8] R. Wang, H. Chen, C. Guan, W. Gong, and Z. Zhang, "Research on the fault monitoring method of marine diesel engines based on the manifold learning and isolation forest," *Appl. Ocean Res.*, vol. 112, Jul. 2021, Art. no. 102681, doi: 10.1016/j.apor.2021.102681.

[9] J. Jeon and G. Theotokatos, "A framework to assure the trustworthiness of physical model-based digital twins for marine engines," *J. Mar. Sci. Eng.*, vol. 12, no. 4, p. 595, Mar. 2024, doi: 10.3390/jmse12040595.

[10] P. Han, A. L. Ellefsen, G. Li, V. Asoy, and H. Zhang, "Fault prognostics using LSTM networks: Application to marine diesel engine," *IEEE Sensors J.*, vol. 21, no. 22, pp. 25986–25994, Nov. 2021, doi: 10.1109/JSEN.2021.3119151.

[11] Y. Tan, H. Tian, R. Jiang, Y. Lin, and J. Zhang, "A comparative investigation of data-driven approaches based on one-class classifiers for condition monitoring of marine machinery system," *Ocean Eng.*, vol. 201, Apr. 2020, Art. no. 107174, doi: 10.1016/j.oceaneng.2020.107174.

[12] M. Cheliotis, I. Lazakis, and G. Theotokatos, "Machine learning and data-driven fault detection for ship systems operations," *Ocean Eng.*, vol. 216, Nov. 2020, Art. no. 107968, doi: 10.1016/j.oceaneng.2020.107968.

[13] C. He, T. Wu, R. Gu, Z. Jin, R. Ma, and H. Qu, "Rolling bearing fault diagnosis based on composite multiscale permutation entropy and reverse cognitive fruit fly optimization algorithm–extreme learning machine," *Measurement*, vol. 173, Mar. 2021, Art. no. 108636, doi: 10.1016/j.measurement.2020.108636.

[14] T. Xie, W. Zhang, Y. Zhang, Z. Ahmed, and Y. Tang, "Marine current turbine multifault diagnosis based on optimization resampled modulus feature and 1-D-CNN," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023, doi: 10.1109/TIM.2023.3267342.

[15] C. Velasco-Gallego and I. Lazakis, "Mar-RUL: A remaining useful life prediction approach for fault prognostics of marine machinery," *Appl. Ocean Res.*, vol. 140, Nov. 2023, Art. no. 103735, doi: 10.1016/j.apor.2023.103735.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.

[16] W. Li et al., "A perspective survey on deep transfer learning for fault diagnosis in industrial scenarios: Theories, applications and challenges," *Mech. Syst. Signal Process.*, vol. 167, Mar. 2022, Art. no. 108487, doi: 10.1016/j.ymssp.2021.108487.

[17] R. Huang, J. Li, Y. Liao, J. Chen, Z. Wang, and W. Li, "Deep adversarial capsule network for compound fault diagnosis of machinery toward multidomain generalization task," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021, doi: 10.1109/TIM.2020.3042300.

[18] X. Zhang, J. Liu, X. Zhang, and Y. Lu, "Self-supervised graph feature enhancement and scale attention for mechanical signal node-level representation and diagnosis," *Adv. Eng. Informat.*, vol. 65, May 2025, Art. no. 103197, doi: 10.1016/j.aei.2025.103197.

[19] Y. Xu et al., "Domain constrained cascadic multireceptive learning networks for machine health monitoring in complex manufacturing systems," *J. Manuf. Syst.*, vol. 80, pp. 563–577, Jun. 2025, doi: 10.1016/j.jmsy.2025.03.021.

[20] D. Li, J. Chen, R. Huang, Z. Chen, and W. Li, "Sensor-aware CapsNet: Towards trustworthy multisensory fusion for remaining useful life prediction," *J. Manuf. Syst.*, vol. 72, pp. 26–37, Feb. 2024, doi: 10.1016/j.jmsy.2023.11.009.

[21] K.-M. Tsitsilonis, G. Theotokatos, C. Patil, and A. Coraddu, "Health assessment framework of marine engines enabled by digital twins," *Int. J. Engine Res.*, vol. 24, no. 7, pp. 3264–3281, Jul. 2023, doi: 10.1177/14680874221146835.

[22] Y. Bo, H. Wu, W. Che, Z. Zhang, X. Li, and L. Myagkov, "Methodology and application of digital twin-driven diesel engine fault diagnosis and virtual fault model acquisition," *Eng. Appl. Artif. Intell.*, vol. 131, May 2024, Art. no. 107853, doi: 10.1016/j.engappai.2024.107853.

[23] Y. Qin, H. Liu, and Y. Mao, "Faulty rolling bearing digital twin model and its application in fault diagnosis with imbalanced samples," *Adv. Eng. Informat.*, vol. 61, Aug. 2024, Art. no. 102513, doi: 10.1016/j.aei.2024.102513.

[24] S. Hautala, M. Mikulski, E. Söderäng, X. Storm, and S. Niemi, "Toward a digital twin of a mid-speed marine engine: From detailed 1D engine model to real-time implementation on a target platform," *Int. J. Engine Res.*, vol. 24, no. 12, pp. 4553–4571, Dec. 2023, doi: 10.1177/14680874221106168.

[25] S. Li and F. Brennan, "Implementation of digital twin-enabled virtually monitored data in inspection planning," *Appl. Ocean Res.*, vol. 144, Mar. 2024, Art. no. 103903.

[26] Z. Mousavi, S. Varahram, M. M. Ettefagh, M. H. Sadeghi, W.-Q. Feng, and M. Bayat, "A digital twin-based framework for damage detection of a floating wind turbine structure under various loading conditions based on deep learning approach," *Ocean Eng.*, vol. 292, Jan. 2024, Art. no. 116563, doi: 10.1016/j.oceaneng.2023.116563.

[27] Y. Huang, J. Tao, G. Sun, T. Wu, L. Yu, and X. Zhao, "A novel digital twin approach based on deep multimodal information fusion for aero-engine fault diagnosis," *Energy*, vol. 270, May 2023, Art. no. 126894, doi: 10.1016/j.energy.2023.126894.

[28] L. Zhou, H. Wang, and S. Xu, "Aero-engine gas path system health assessment based on depth digital twin," *Eng. Failure Anal.*, vol. 142, Dec. 2022, Art. no. 106790.

[29] J. Zhang, Z. Wang, S. Li, and P. Wei, "A digital twin approach for gas turbine performance based on deep multi-model fusion," *Appl. Thermal Eng.*, vol. 246, Jun. 2024, Art. no. 122954, doi: 10.1016/j.applthermaleng.2024.122954.

[30] G. Theotokatos, S. Stoumpos, V. Bolbot, and E. Boulougouris, "Simulation-based investigation of a marine dual-fuel engine," *J. Mar. Eng. Technol.*, vol. 19, no. 1, pp. 5–16, Jan. 2020, doi: 10.1080/20464177.2020.1717266.

[31] C. W. Hong and J. Kim, "Exhaust temperature prediction for gas turbine performance estimation by using deep learning," *J. Electr. Eng. Technol.*, vol. 18, no. 4, pp. 3117–3125, Jul. 2023, doi: 10.1007/s42835-023-01488-x.

[32] S. Chu, J. Zhang, F. Liu, X. Kong, Z. Jiang, and Z. Mao, "Fault identification model of diesel engine based on mixed attention: Single-cylinder fault data driven whole-cylinder diagnosis," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124769, doi: 10.1016/j.eswa.2024.124769.

[33] B. Hu, J. Liu, and Y. Xu, "A novel multi-scale convolutional neural network incorporating multiple attention mechanisms for bearing fault diagnosis," *Measurement*, vol. 242, Jan. 2025, Art. no. 115927, doi: 10.1016/j.measurement.2024.115927.

[34] W. Chen, D. Zhang, M. Li, and D.-J. Lee, "STCAM: Spatial–temporal and channel attention module for dynamic facial expression recognition," *IEEE Trans. Affect. Comput.*, vol. 14, no. 1, pp. 800–810, Jan. 2023, doi: 10.1109/TAFFC.2020.3027340.

[35] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020, doi: 10.1109/TPAMI.2019.2913372.

[36] Y. Dong, H. Jiang, Z. Wu, Q. Yang, and Y. Liu, "Digital twin-assisted multiscale residual-self-attention feature fusion network for hypersonic flight vehicle fault diagnosis," *Rel. Eng. Syst. Saf.*, vol. 235, Jul. 2023, Art. no. 109253, doi: 10.1016/j.ress.2023.109253.

[37] H. Tian, L. Yang, and B. Ju, "Spatial correlation and temporal attention-based LSTM for remaining useful life prediction of turbofan engine," *Measurement*, vol. 214, Jun. 2023, Art. no. 112816, doi: 10.1016/j.measurement.2023.112816.

[38] Z. Liu, Y. Shen, V. B. Lakshminarasimhan, P. P. Liang, A. B. Zadeh, and L.-P. Morency, "Efficient low-rank multimodal fusion with modality-specific factors," in *Proc. 56th Annu. Meeting Assoc. Comput. Linguistics*, 2018, pp. 2247–2256.

[39] X. Xu, Y. Lin, and C. Ye, "Fault diagnosis of marine machinery via an intelligent data-driven framework," *Ocean Eng.*, vol. 289, Dec. 2023, Art. no. 116302, doi: 10.1016/j.oceaneng.2023.116302.

[40] Z. Liao et al., "Uncertainties in gas-path diagnosis of gas turbines: Representation and impact analysis," *Aerosp. Sci. Technol.*, vol. 113, Jun. 2021, Art. no. 106724, doi: 10.1016/j.ast.2021.106724.

[41] R. Zhou, J. Cao, G. Zhang, X. Yang, and X. Wang, "Heat load forecasting of marine diesel engine based on long short-term memory network," *Appl. Sci.*, vol. 13, no. 2, p. 1099, Jan. 2023, doi: 10.3390/app13021099.

[42] C. Velasco-Gallego and I. Lazakis, "RADIS: A real-time anomaly detection intelligent system for fault diagnosis of marine machinery," *Expert Syst. Appl.*, vol. 204, Oct. 2022, Art. no. 117634, doi: 10.1016/j.eswa.2022.117634.

[43] P. Pan, D. Zhao, and Y. Li, "A fault diagnosis framework for rotating machinery of marine equipment: A semi-supervised learning framework based on contractive stacked autoencoder," *Proc. Inst. Mech. Eng., M, J. Eng. Maritime Environ.*, vol. 237, no. 3, pp. 625–636, Aug. 2023, doi: 10.1177/14750902221143827.

[44] S. Rigas, P. Tzouveli, and S. Kollias, "An end-to-end deep learning framework for fault detection in marine machinery," *Sensors*, vol. 24, no. 16, p. 5310, Aug. 2024, doi: 10.3390/s24165310.



Jiawen Sun received the B.Sc. degree in geographic information science from Dalian Maritime University, Dalian, China, in 2018, where he is currently pursuing the Ph.D. degree in traffic and transportation engineering.

His current research interests include machine learning, data-driven fault prediction methods, and digital twin ships.



Hongxiang Ren received the Ph.D. degree in traffic and transportation engineering from Dalian Maritime University, Dalian, China, in 2009.

He is currently a Professor with Dalian Maritime University. His current research interests include virtual reality, maritime simulation, and intelligent ships.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.



Yaxin Dong received the M.Sc. degree in traffic information engineering and control from Dalian Maritime University, Dalian, China, in 2020, where she is currently pursuing the Ph.D. degree in traffic information engineering and control.

Her current research interests include maritime traffic safety and reliability analysis.



Boxiang Zhang received the M.Sc. degree in traffic information engineering and control from Dalian Maritime University, Dalian, China, in 2024, where he is currently pursuing the Ph.D. degree in traffic information engineering and control.

His current research interests include maritime simulation, machine learning, and numerical modeling.



Shaoyang Qiu received the Ph.D. degree from the Key Laboratory of Marine Dynamic Simulation and Control, Ministry of Communications, Dalian Maritime University, Dalian, China, in 2021.

He is currently a Lecturer with the College of Artificial Intelligence, Dalian Maritime University. He has authored or co-authored more than ten papers on modeling and simulation. He has conducted in-depth research on the motion modeling of ship lifeboats and rescue boats and has applied these models to training simulators. His current research interests include digital twins, ship motion modeling, and virtual reality.



Yi Zhou received the B.Sc. degree in process equipment and control engineering from Jiangsu Ocean University, Lianyungang, China, in 2008.

He is currently a Senior Engineer with Offshore Oil Engineering Service Company, Tianjin, China. His current research interests include intelligent ships and navigation condition simulation.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on July 04,2026 at 10:16:30 UTC from IEEE Xplore. Restrictions apply.